PocketSmart AI

Epic 5: Testing and Deployment

Objective: Test major components and prepare PocketSmart AI for deployment and final submission.

Files and Folders

Project structure

```
PocketSmart-AI/
■■■ tests/
■   ■■■ __init__.py
■   ■■■ test_app.py
■■■ README.md
■■■ requirements.txt
■■■ .env.example
■■■ .gitignore
■■■ app/
    ■■■ main.py
    ■■■ auth.py
    ■■■ database.py
    ■■■ models.py
    ■■■ schemas.py
    ■■■ dependencies.py
    ■■■ routers/
    ■■■ services/
    ■■■ templates/
    ■■■ static/
```

Testing

- tests/test_app.py
- Health endpoint
- Home page
- Home planner
- Party planner
- User registration
- User login
- Authentication cookie

Application Testing

- Application startup
- Database connection
- User registration
- User login/logout
- JWT authentication
- Home Planner
- Party Planner
- Jewelry Planner
- Budget validation
- Recommendation generation
- Recommendation history

- Image upload
- Gemini integration
- Fallback recommendation system
- Frontend navigation
- Responsive UI

Deployment Configuration

- .env.example — Application and Gemini environment configuration.
- requirements.txt — Python dependencies.
- .gitignore — Excludes environment, cache and generated files.

Run Commands

Install dependencies

```
pip install -r requirements.txt
```

Run application

```
python -m uvicorn app.main:app --reload
```

Run tests

```
python -m pytest -q
```

Application

```
http://127.0.0.1:8000
```

API documentation

```
http://127.0.0.1:8000/docs
```

Epic 5 Deliverable

Tested and runnable PocketSmart AI application ready for final demonstration and deployment.